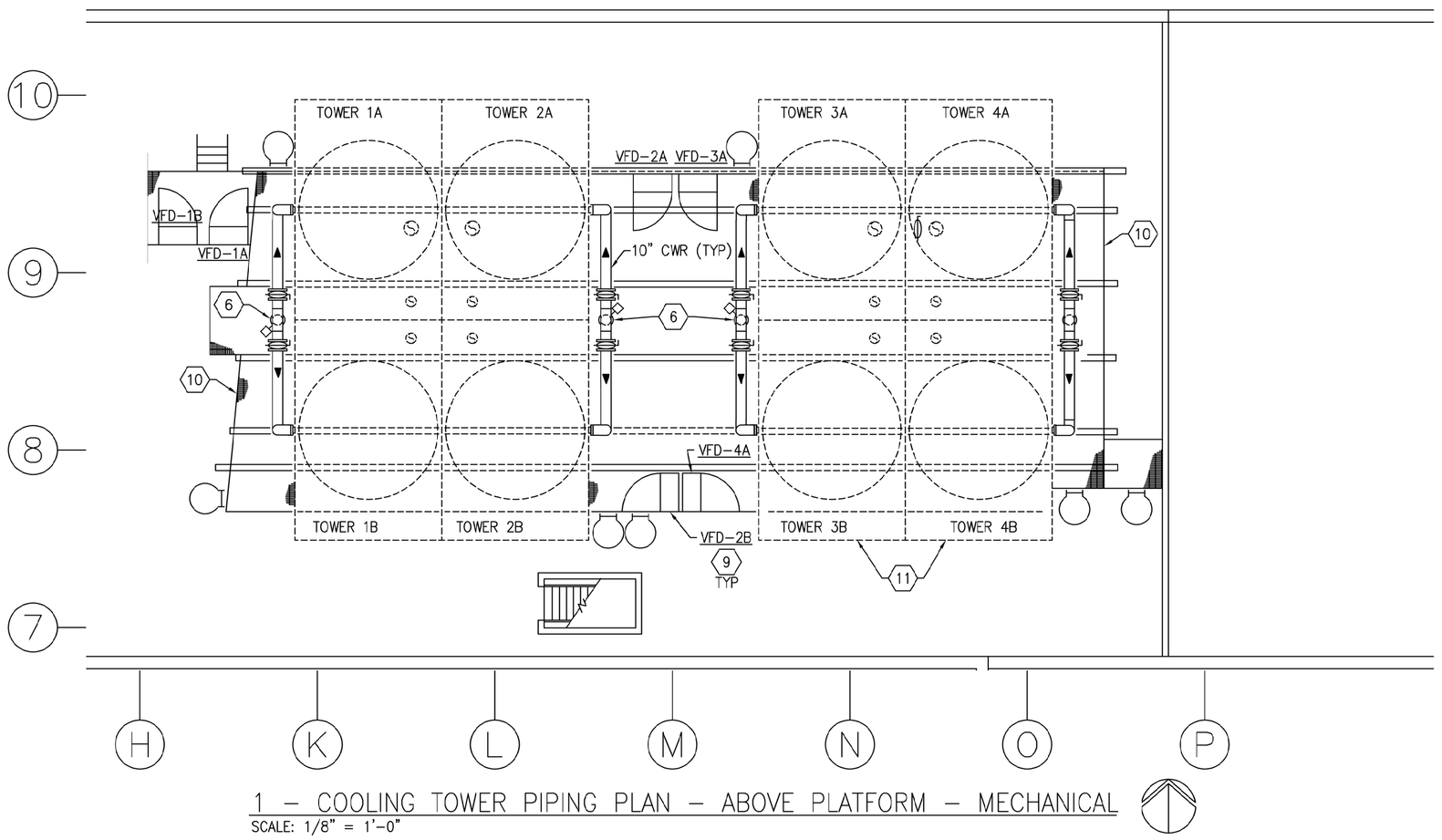
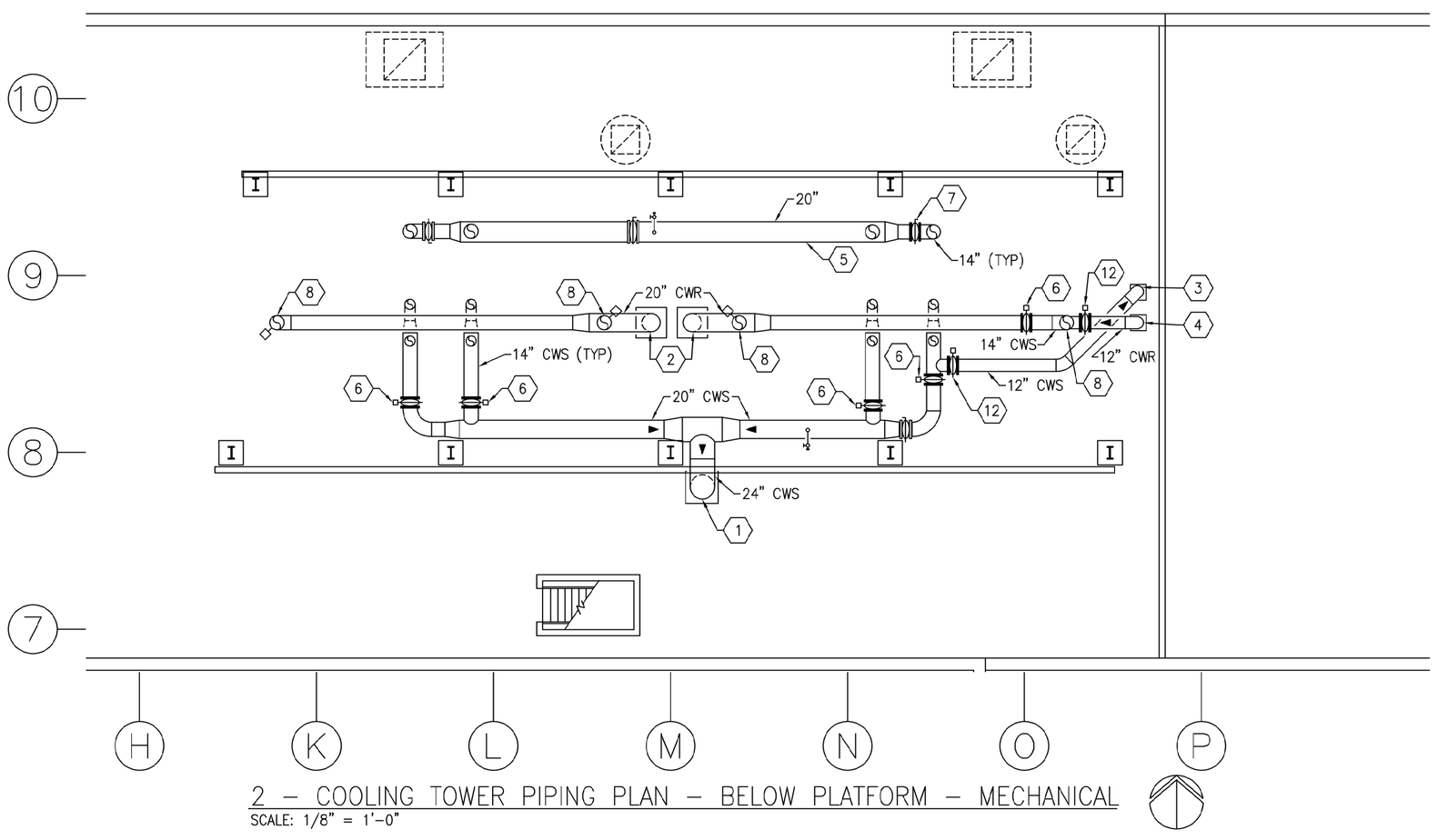


1 – COOLING TOWER PIPING PLAN – ABOVE PLATFORM – MECHANICAL

SCALE: 1/8" = 1'-0"



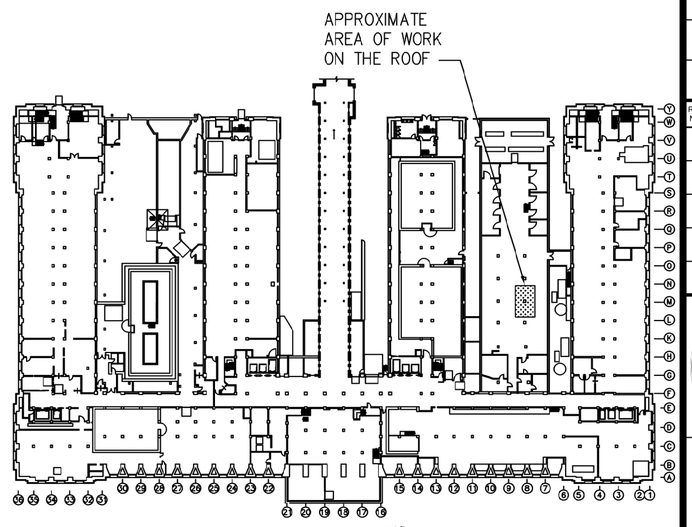
2 – COOLING TOWER PIPING PLAN – BELOW PLATFORM – MECHANICAL

SCALE: 1/8" = 1'-0"

ALL EIGHT–14" VALVES CONTROLLING COOLING TOWERS ARE NO.
TWO–12" VALVES CONTROLLING HEAT–EX ARE NC.

(#) M–4.1 AND M–4.2 NOTES:

1. EXISTING CONDENSER WATER SUPPLY (CWS) PIPE DOWN TO CHILLERS.
2. EXISTING CONDENSER WATER RETURN (CWR) PIPE FROM CHILLERS.
3. EXISTING CONDENSER WATER SUPPLY PIPE DOWN TO HEAT EXCHANGER.
4. EXISTING CONDENSER WATER RETURN FROM HEAT EXCHANGER.
5. EXISTING EQUALIZING PIPE.
6. REPLACE THE EXISTING 14" CONTROL VALVE WITH BRAY SERIES 31–119 BUTTERFLY VALVE AND SERIES 93–2106 PNEUMATIC ACTUATOR, SPRING RETURN AND FAIL CLOSE. CAST IRON VALVE BODY WITH DUCTILE IRON NYLON COATED DISC, EDPM SEAT.
7. REPLACE THE EXISTING 14" CONTROL VALVE WITH BRAY SERIES 31–119 BUTTERFLY VALVE WITH HAND WHEEL GEAR OPERATOR. CAST IRON VALVE BODY WITH DUCTILE IRON NYLON COATED DISC, EDPM SEAT.
8. EXISTING CONDENSER WATER RETURN PIPE UP TO COOLING TOWER ABOVE.
9. FURNISH 480 VAC, 60 HP VFD. EATON CFX06034AACJRARH WITH CFX NEMA 3R ENCLOSURE, LOW OVERLOAD, MAHUAL HOA BYPASS, REDUCED VOLTAGE STARTER FOR BYPASS–SOFT STOP WITH BACNET. (INSTALLATION BY ELECTRICAL CONTRACTOR). REFER TO ELECTRICAL DRAWINGS FOR ADDITIONAL REQUIREMENTS.
10. EXISTING COOLING TOWER PLATFORM.
11. REPLACE THE EXISTING 60 HP VFD IN THE MECHANICAL ROOM WITH NEW 480 VAC, 60 HP VFD. EATON CFX06034AACJRARH WITH CFX NEMA 12 ENCLOSURE, LOW OVERLOAD, MAHUAL HOA BYPASS, REDUCED VOLTAGE STARTER FOR BYPASS–SOFT STOP WITH BACNET. (INSTALLATION BY ELECTRICAL CONTRACTOR).
12. REPLACE THE EXISTING 12" CONTROL VALVE WITH BRAY SERIES 31–119 BUTTERFLY VALVE AND SERIES 93–2106 PNEUMATIC ACTUATOR, SPRING RETURN AND FAIL CLOSE. CAST IRON VALVE BODY WITH DUCTILE IRON NYLON COATED DISC, EDPM SEAT.



KEY PLAN

SCALE: NONE

ASI
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SEAL:

OFFICE	SIGNATURE	DATE
Engineer, Chief		
Engineer, Division Manager		
Engineer, COTR		
Engineer, Electrical		
Engineer, Mechanical		
Engineer, Structural		
Environmental		
Safety		
Security		
Information Systems		
Requesting Office Chief		
Requesting Office POC		

REVISION NUMBER	REMARKS	APPROVAL	DATE
1	PRELIM. CONTRACTOR REVIEW		05/25/12
2	CONSTRUCTION SET		06/08/12
3	FINAL CONSTRUCTION SET		10/09/12
4			
5			

DEPARTMENT OF THE TREASURY
BUREAU OF ENGRAVING AND PRINTING

COOLING TOWER PIPING PLANS - MECHANICAL

A48.12010

DRAWN BY: BS DATE:
CHECKED BY: APH DATE:
DATE OF ISSUE:
SHEET OF
SCALE: SEE PLAN

DWG. NO. M-4